SMART INDIA HACKATHON 2026

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SentinelX

- **Problem Statement ID - SIH26188**
- **Problem Statement Title - AI-Based Fake Identity & Document Screening System(SSB)**
- **Theme - Miscellaneous**
- **PS Category - Software**
- **Team Name - SentinelX**

SentinelX

SentinelX – AI-Based Fake Identity & Document Screening System

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Proposed Solution & How it addresses the problem

- AI-powered mobile application for border security personnel.
- Instantly detects fake or tampered identity documents.
- Provides real-time verification results.
- Generates security alerts for suspicious documents.
- Reduces manual verification time.
- Improves accuracy in document screening.
- Helps prevent identity fraud.
- Supports faster border clearance.



Innovation & Uniqueness

- AI-based forgery detection.
- On-device real-time verification.
- Officer-friendly interface.
- Secure verification history.

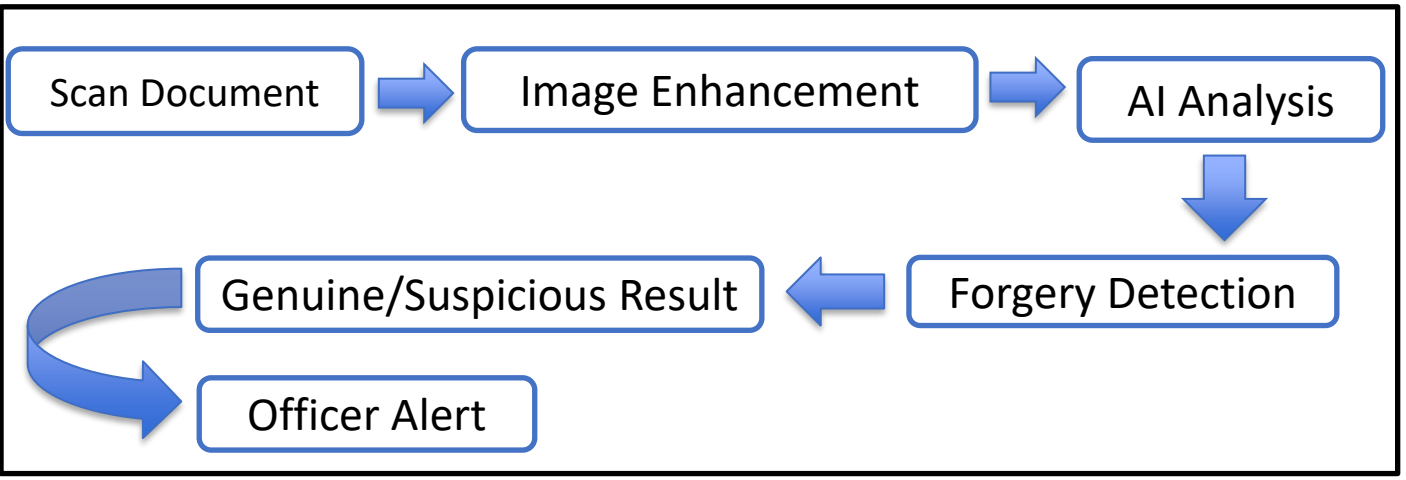
TECHNICAL APPROACH

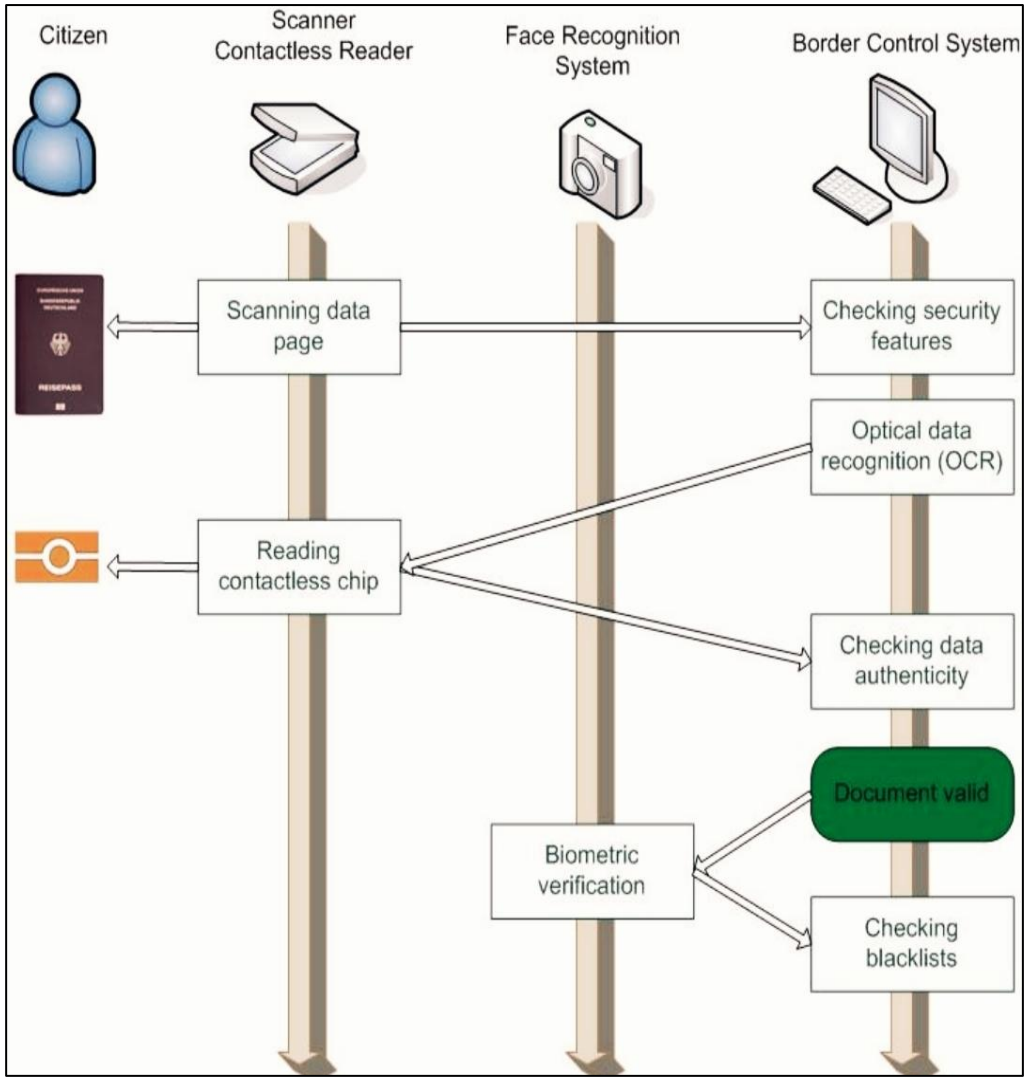
Technology Used

- Kotlin (Android)
- TensorFlow Lite
- Firebase
- Computer Vision
- QR/Barcode Validation

Methodology & Process

Flowchart:





FEASIBILITY AND VIABILITY

Feasibility

- Android-based deployment
- AI runs on edge devices
- Works offline with periodic sync
- Scalable for multiple checkpoints

Potential Challenges

- Different document formats
- Low-quality images
- Advanced forgery techniques
- Poor lighting conditions

Strategies too overcome

- Image enhancement
- Multi-layer AI checks
- QR/Barcode verification
- Continuous model improvement

Work Flow Diagram



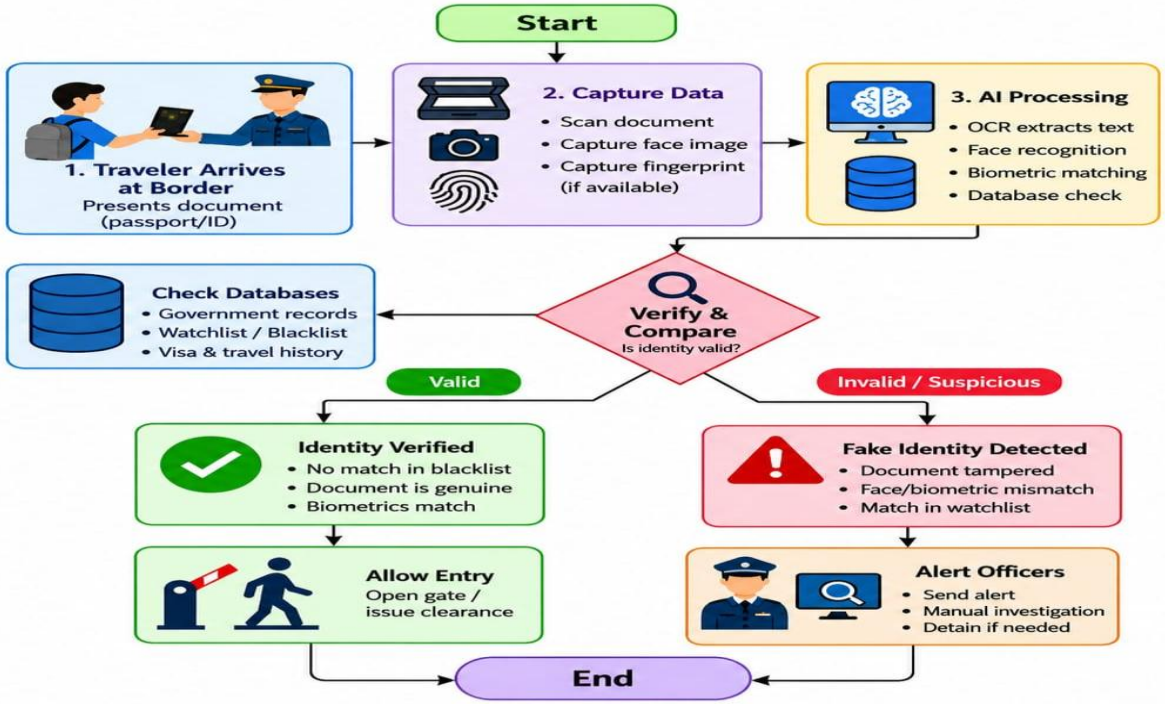
IMPACT AND BENEFITS

Potential Impact

- Strengthens national security
- Faster border verification
- Reduced manual/officer workload
- Better detection of fake identities

Benefits

- Detects fake identities quickly
- Reduces human errors
- Saves verification time
- Improves officer efficiency





RESEARCH AND REFERENCES



References

- Smart India Hackathon 2026 – Official Problem Statement SIH26188.
- TensorFlow Lite Documentation.
- Google ML Kit Documentation.
- Firebase Documentation.
- Research papers on AI-based document forgery detection.
- Open-source datasets for identity document analysis.